Introduction to Theoretical Computer Science

CS2490 - Session 19: Pumping Lemma

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Recall: Pumping Lemma

Theorem

Let Σ be an alphabet. If L is a regular language over Σ, then there exists an integer $p \geq 1$ (the pumping length) such that, for every word $w \in L$ with $|w| \geq p$, there exists a factorization $w = xyz$ with $x, y, z \in \Sigma^*$ such that:

- $|y| \geq 1$,
- $|xy| \leq p$,
- *for all $i \geq 0$, the word $xy^i z$ is in L.*

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- We administer the *coup de grâce* by finding an i such that $xy^i z \notin L$.

More precisely, we design a winning strategy to beat the opponent's moves, whatever they may be.

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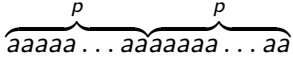
Question: Could we have chosen $w = a^p a^p$ instead?

Example 1 (cont'd)

Answer: NO!

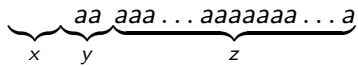
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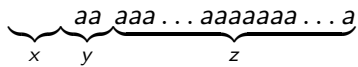
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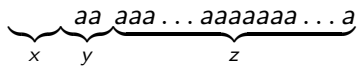
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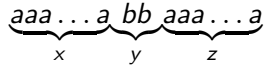
The universe wins! Our first move were not strong enough.

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$$\underbrace{aaa\ldots a}_{x}\ \underbrace{bb}_{y}\ \underbrace{aaa\ldots a}_{z}$$

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The universe cannot do this: the factorization must have $|xy| \leq p$. Our first move is strong enough!

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Question: Could we have pumped 0 times, like for the two other examples?

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Hence, y is either of the form $(ab)^k$ for some $k \geq 1$, or $y = a(ba)^k$ for some $k \geq 0$, or $y = b(ab)^k$ for some $k \geq 0$, or $y = (ba)^k$ for some $k \geq 1$.

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In the fourth case, pump 0 times: $xy^0z = xz = a(ba)^{p-1-k}ba^{p-1}$ is supposedly in L. But since $p - k \leq p - 1$, we have $xz \notin L$.

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Question: Our reasoning assumes $p \geq 2$. Can we do that?

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Answer: YES!

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Remember that the p given by the opponent is the number of states of a DFA supposedly recognizing L. If our opponent chooses $p = 1$, then we can easily disprove that their DFA recognizes L since the only DFA with 1 states recognize either $\emptyset$ or Σ^*.

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More training to come!